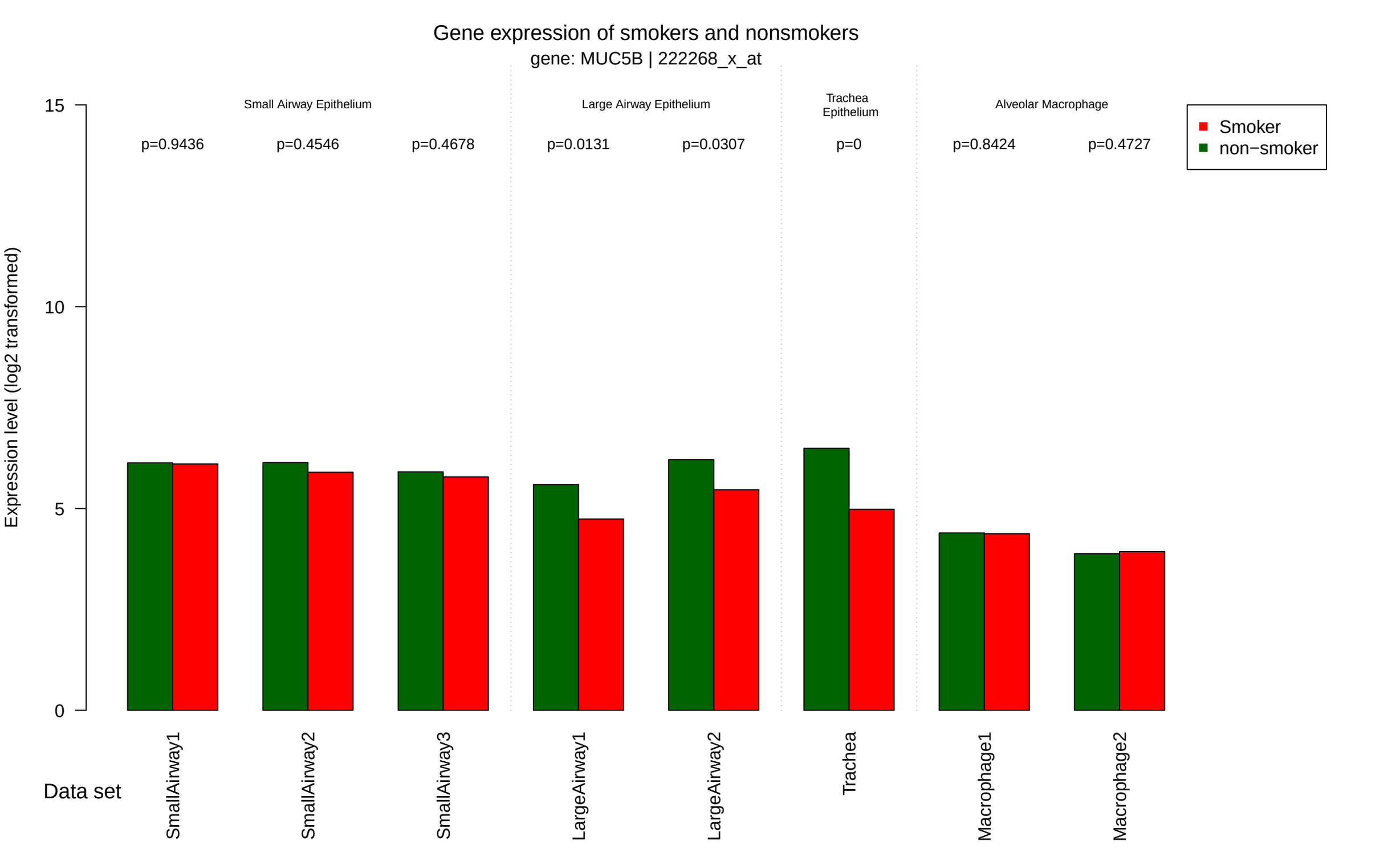
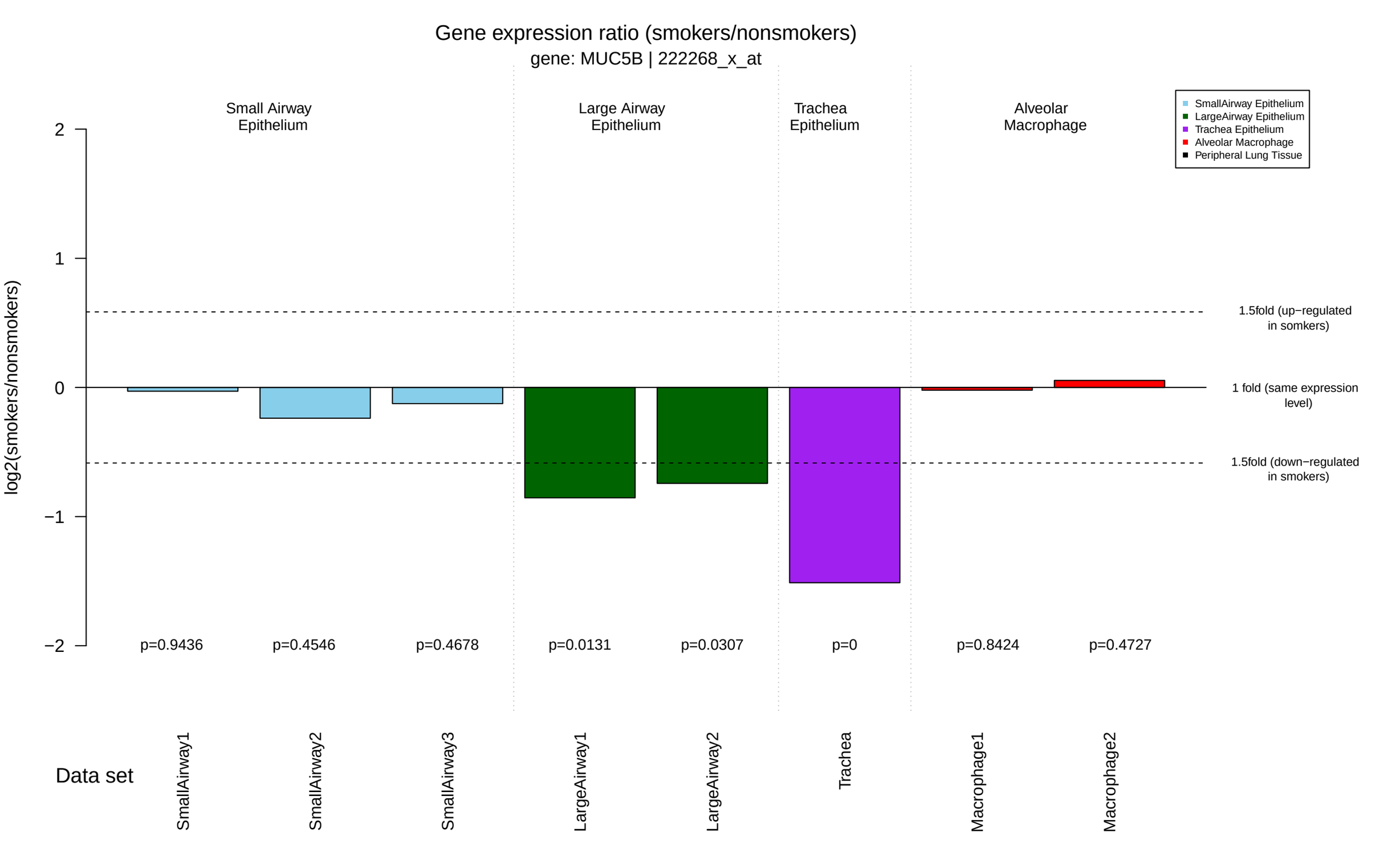
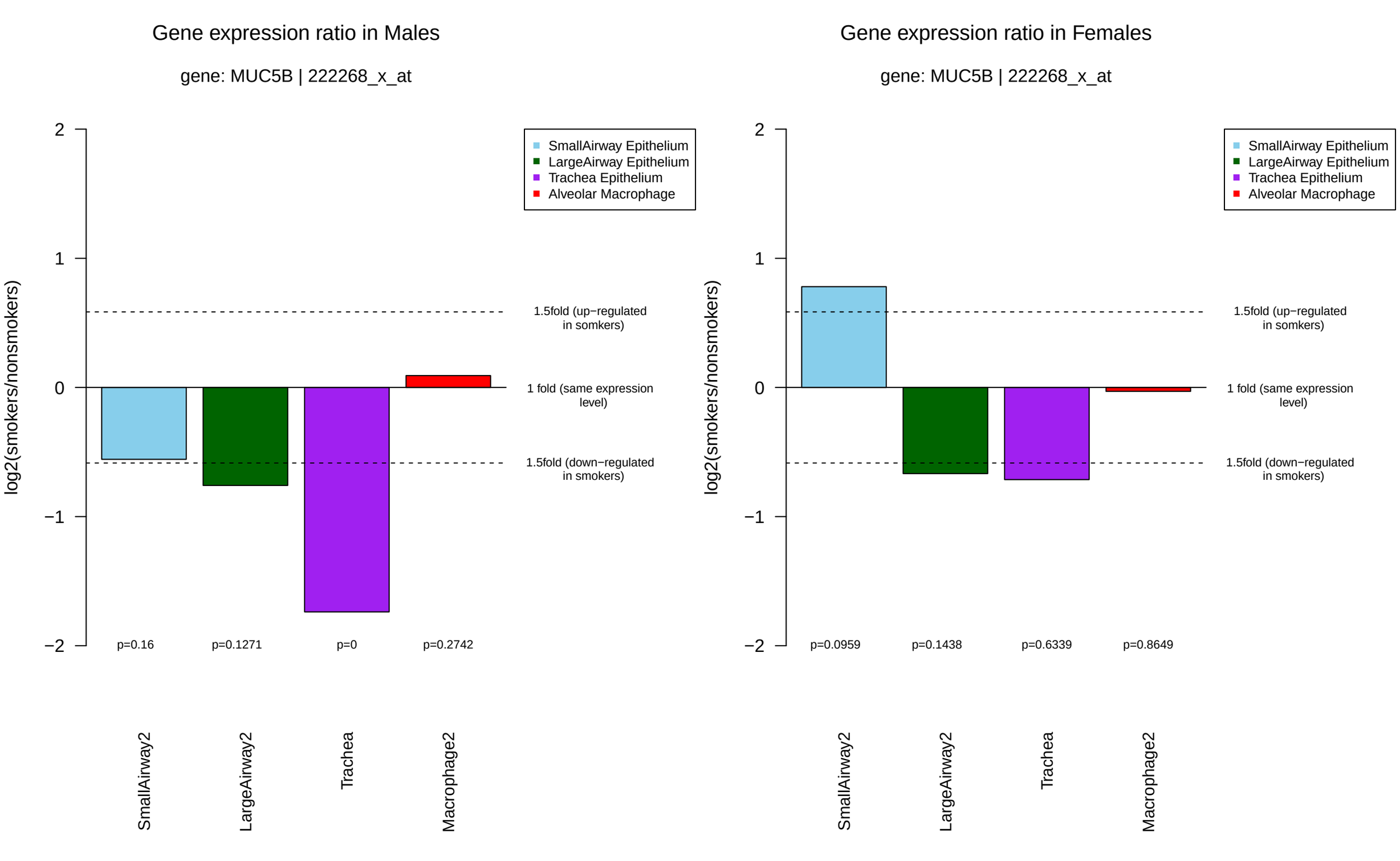
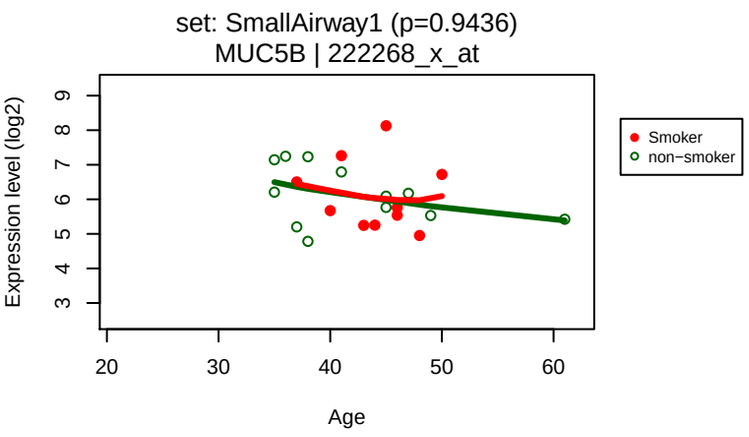
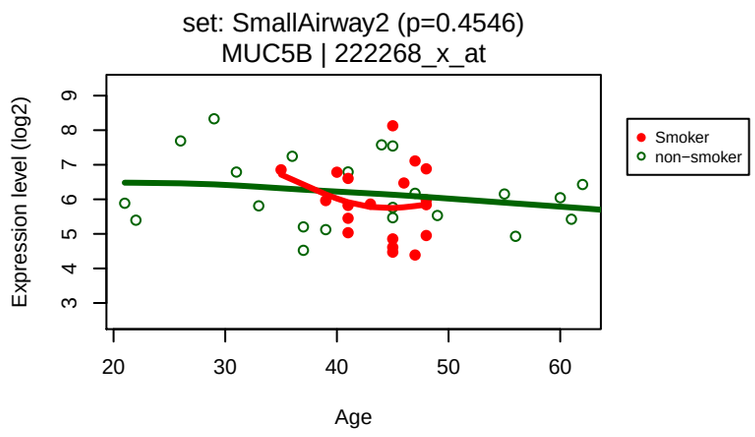
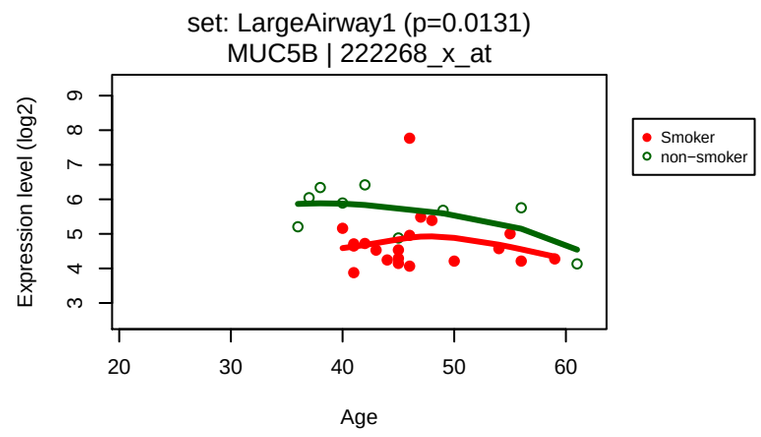

Gene expression of smokers and nonsmokers
gene: MUC5B | 222268_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9436
p=0.4546
p=0.4678
p=0.0131
p=0.0307
p=0
p=0.8424
p=0.4727
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

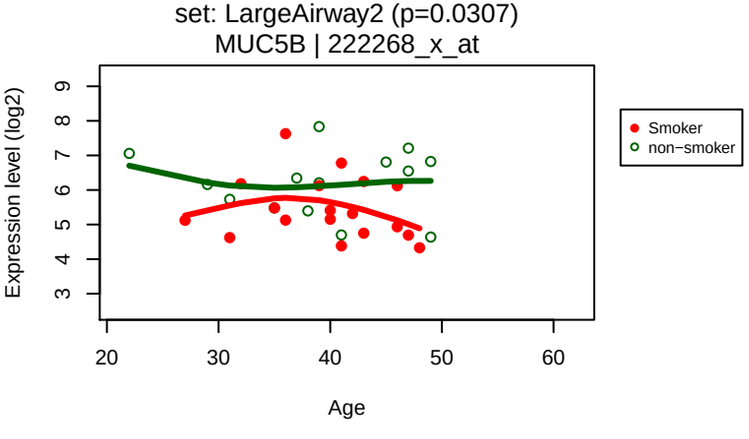

Gene expression ratio (smokers/nonsmokers)
gene: MUC5B | 222268_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9436
p=0.4546
p=0.4678
p=0.0131
p=0.0307
p=0
p=0.8424
p=0.4727
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

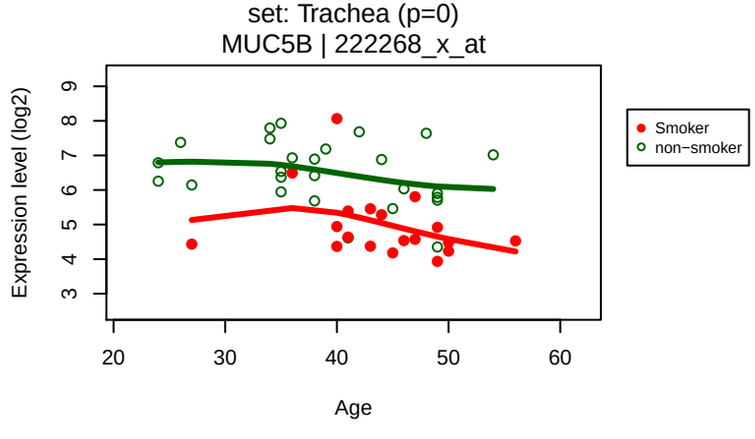

Gene expression ratio in Males
gene: MUC5B | 222268_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.16
p=0.1271
p=0
p=0.2742
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC5B | 222268_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0959
p=0.1438
p=0.6339
p=0.8649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

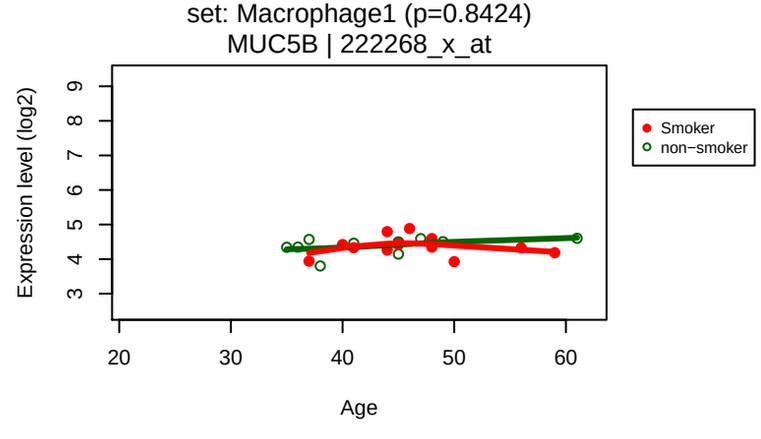

set: SmallAirway1 (p=0.9436)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker

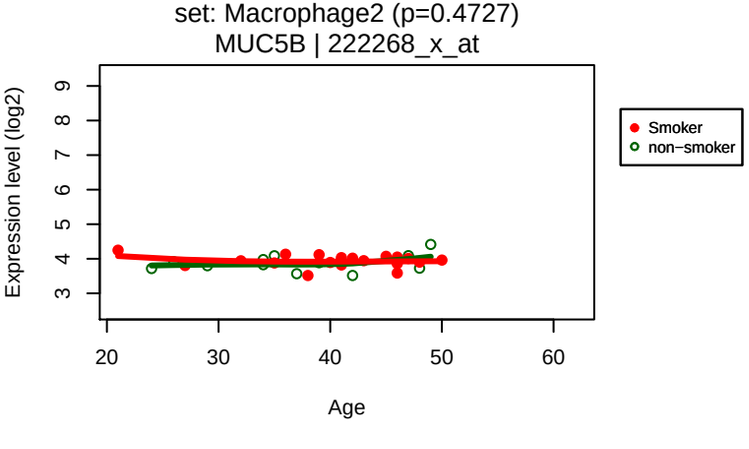

set: SmallAirway2 (p=0.4546)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker

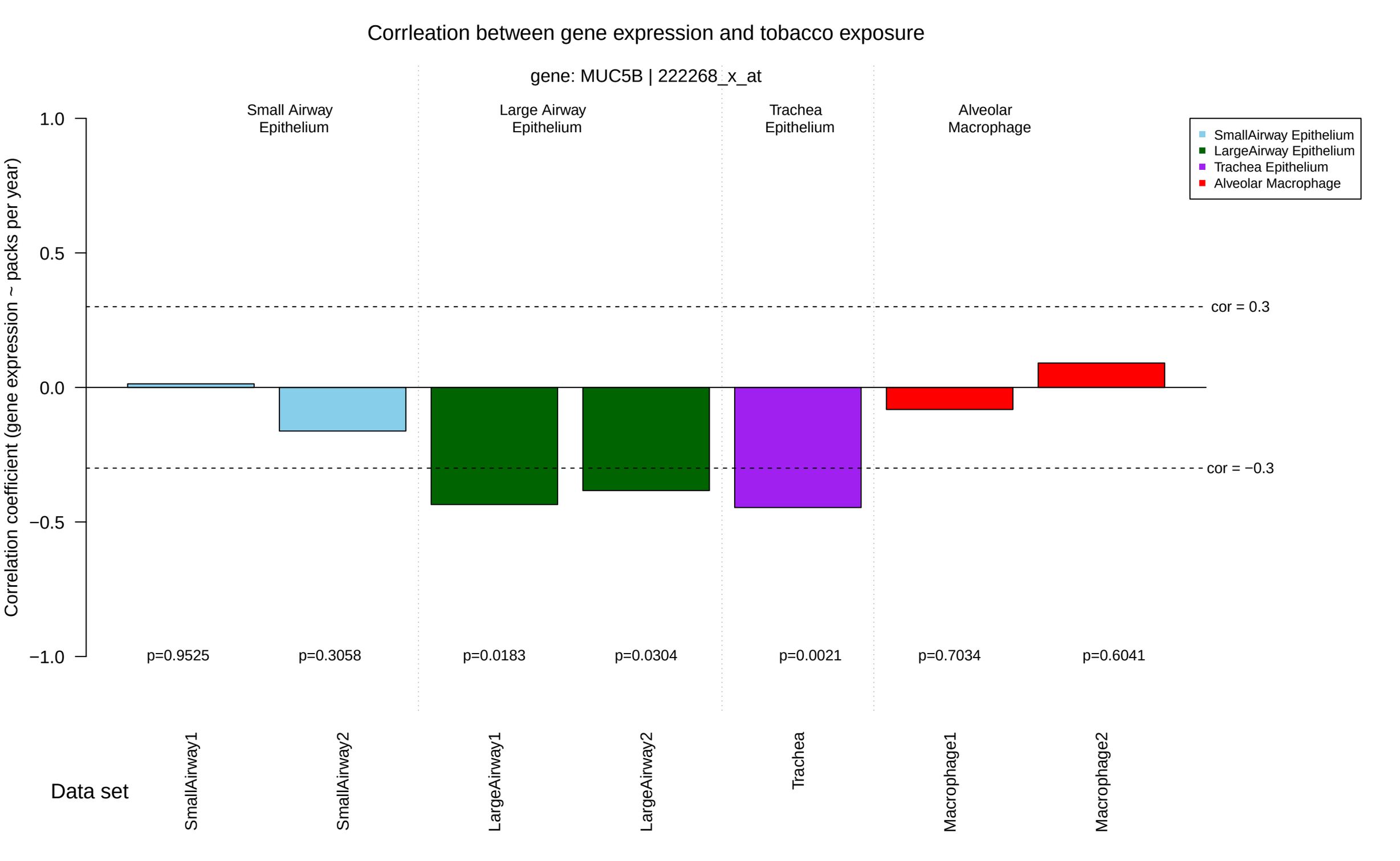

set: LargeAirway1 (p=0.0131)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0307)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8424)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4727)
MUC5B | 222268_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC5B | 222268_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9525
p=0.3058
p=0.0183
p=0.0304
p=0.0021
p=0.7034
p=0.6041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2